APT: APT is best for Debian platforms and it takes care of all the dependencies as well.

Add the following to */etc/apt/sources.list*

```
deb http://debian.saltstack.com/debian squeeze-saltstack main
deb http://backports.debian.org/debian-backports squeeze-backports main contrib non-free
```

And import the required key before installing the following:

```
wget -q -O- "http://debian.saltstack.com/debian-salt-team-joehealy.gpg.key" | apt-key add -
apt-get install salt-master
apt-get install salt-minion
```

TAR: This is the least preferred way because of the dependencies involved, but is do-able in the following way:

```
wget https://pypi.python.org/packages/source/s/salt/salt-2014.1.10.tar.gz --no-check-certificate
tar -zxvf salt-2014.1.10.tar.gz
cd salt-2014.1.10
python26 setup.py install
wget https://bootstrap.pypa.io/get-pip.py --no-check-certificate (for pip command for python26)
```

Dependencies for TAR: Using PIP or Source Tar, the following packages have to be installed for SaltStack and Python to go hand-in-hand:

```
yum install python26-jinja2.noarch python26-PyYAML python26-zmq.x86_64 python26-m2crypto.x86_64 python26-msgpack.x86_64 python26-crypto.x86_64 python26-PyYAML.x86_64
/usr/bin/pip2.6 install msgpack-pure
salt-master --versions-report

           Salt: 2014.1.10
         Python: 2.6.8 (unknown, Nov  7 2012, 14:47:45)
         Jinja2: 2.5.5
       M2Crypto: 0.21.1
 msgpack-python: 0.1.12
   msgpack-pure: 0.1.3
       pycrypto: 2.3
         PyYAML: 3.08
          PyZMQ: 2.1.9
            ZMQ: 2.1.9
```

Configuration

Master:

- If it is a tar-based install, create the folder */etc/salt* and copy the file *conf/master* from the *Salt install* folder to */etc/salt/master*. In other cases, the file should be there by default.
- By default, it will take the path */srv/salt* and */srv/pillar* as the base folders, if not mentioned. But still, we can go and uncomment the following lines in */etc/salt/master*
  - file_roots:
  - base:
  - - /srv/salt
- Run the command *salt-master* with a *-d* if you want to run it as a background daemon.

Client:

- If it is a tar-based install, create the folder */etc/salt* and copy the file *conf/minion* from the *salt install* folder to */etc/salt/minion.* In other cases, the file should be there, by default.
- Add *<ip-address> salt salt.master.com* to */etc/hosts*. Only and only if DNS doesn't have entry for the Salt master, do we need to add an entry here that resolves entries to the word 'salt'.
- Search for *"#master:"* in the file */etc/salt/minion;* uncomment it before adding the name of the Salt master after the colon and saving the file.
- Run the command *salt-minion* with a *-d* if you want to run it as a background daemon. This will send an authorisation request to the master.

Activation:

- From the server, list the machines waiting for authorisation using:
  - salt-key -L
- To authorise a node, sign the respective node
  - salt-key -a 'node-name'
  - salt-key -a (accepts all nodes)

Testing:

- Run a test from the master to the client.
  - salt '*' test.ping
  - which gives output as…
    - Node_name:
    - True

Uninstalling:

```
yum remove salt salt-master salt-minion (RedHat or SuSE)
zypper remove salt salt-master salt-minion (SuSE)
apt-get autoremove salt salt-master salt-minion (Debian, Ubuntu)
```

If we haven't removed the untarred *salt-tar* folder, then a *make clean* must clean things up.

Resources

[1] *http://salt.readthedocs.org/en/v0.9.2/topics/download.html*
[2] *http://docs.saltstack.com/en/latest/*
[3] *http://www.linuxjournal.com/content/getting-started-salt-stack-other-configuration-management-system-built-python*

By: Bejoy Abraham Mathews and Davana Satish Gongadi

Bejoy Abraham Mathews works with Servista Inc. He has earlier authored articles on Puppet and is RHCE, the cloud and Big Data certified. He can be contacted at *bejoy.abraham@gmail.com*

Davana Satish Gongadi works with Quadrant 4 System Corporation. He is SCJP certified and is an expert in Java Application Server environments on GNU/Linux. He can be contacted at *gd_satish@yahoo.com.*